

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE CODE: CSA51

COURSE NAME: Cryptography and Network Security

COURSE OUTCOME COVERED

CO2: Analyze Public Key crypto system strategies using RSA and key Exchange for Encryption algorithm to strengthen information security. (BL4,BL5)

Analytical Problem Solving: 40%

QUESTIONS:

Total Marks: 50

Annexure A

Code of Conduct:

*I Eddu Suchier/192524070 certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:

Annexure A Analytical Problem Solving

1. RSA Encryption Analysis

$p=11$

$q=13$

(a) Compute $\phi(n)$

1. Compute n :

$$n = p \times q = 11 \times 13 = 143$$

2. Compute Euler's Totient:

$$\phi(n) = (p - 1)(q - 1)$$

$$\phi(n) = (11 - 1)(13 - 1) = 10 \times 12 = 120$$

- $n = 143$
- $\phi(n) = 120$

(b) Choose $e = 7$ and find the private key d

The private key satisfies:

$$d \times e \equiv 1 \pmod{\phi(n)}$$

$$d \times 7 \equiv 1 \pmod{120}$$

Using the Extended Euclidean Algorithm:

$$7 \times 103 = 721$$

$$721 \bmod 120 = 1$$

Therefore,

$$d = 103$$

- Public Key = **(7, 143)**
- Private Key = **(103, 143)**

(c) Encrypt the message $M = 9$

RSA encryption:

$$C = M^e \bmod n$$

$$C = 9^7 \bmod 143$$

Calculate:

$$9^2 = 81$$

$$9^4 = 81^2 = 6561 \equiv 126 \pmod{143}$$

$$9^7 = 9^4 \times 9^2 \times 9$$

$$= 126 \times 81 \times 9$$

$$126 \times 81 = 10206 \equiv 53 \pmod{143}$$

$$53 \times 9 = 477 \equiv 48 \pmod{143}$$

Ciphertext:

$$\boxed{C = 48}$$

(d) Decrypt the ciphertext

RSA decryption:

$$M = C^d \bmod n$$

$$M = 48^{103} 143$$

Using modular exponentiation,

$$48^{103} 143 = 9$$

Recovered message:

$$\boxed{M = 9}$$

The decrypted message matches the original message, so the encryption and decryption are verified.

(e) Analyze how changing e affects security

- The value of e must be relatively prime to $\phi(n)$.
- A **small** e (such as 3 or 7) speeds up encryption but may be vulnerable if padding is not used properly.
- A **larger** e increases encryption time but can provide better resistance against certain attacks.
- The commonly used value is **65537**, as it offers a good balance between efficiency and security.
- Regardless of e , the overall security of RSA mainly depends on the difficulty of factoring the modulus n .

2. Block Cipher Mode Evaluation

Given:

- Plaintext blocks:
 - $P_1 = 1010$
 - $P_2 = 1010$
 - $P_3 = 1111$
 - $P_4 = 1010$
- Key: $K = 1100$
- IV = 0000 (for CBC)

$$E(P) = P \oplus K$$

(a) Encrypt using ECB and CBC Modes

ECB Mode

Each plaintext block is encrypted independently.

$$C_i = P_i \oplus K$$

Plaintext Key Ciphertext

1010	1100 0110
1010	1100 0110
1111	1100 0011
1010	1100 0110

ECB Ciphertext:

$$\boxed{0110, 0110, 0011, 0110}$$

CBC Mode (IV = 0000)

Formula:

$$C_i = (P_i \oplus C_{i-1}) \oplus K$$

where $C_0 = IV = 0000$

Block 1

$$P_1 \oplus IV = 1010 \oplus 0000 = 1010$$

$$C_1 = 1010 \oplus 1100 = 0110$$

Block 2

$$P_2 \oplus C_1 = 1010 \oplus 0110 = 1100$$

$$C_2 = 1100 \oplus 1100 = 0000$$

Block 3

$$P_3 \oplus C_2 = 1111 \oplus 0000 = 1111$$

$$C_3 = 1111 \oplus 1100 = 0011$$

Block 4

$$P_4 \oplus C_3 = 1010 \oplus 0011 = 1001$$

$$C_4 = 1001 \oplus 1100 = 0101$$

CBC Ciphertext:

0110, 0000, 0011, 0101

(b) Compare the Ciphertext Outputs**Block Plaintext ECB CBC**

1	1010	0110 0110
2	1010	0110 0000
3	1111	0011 0011
4	1010	0110 0101

Comparison:

- In **ECB**, identical plaintext blocks produce identical ciphertext blocks.
- In **CBC**, identical plaintext blocks produce different ciphertext blocks because each block depends on the previous ciphertext.

(c) Which Mode Hides Patterns Better?**CBC mode hides patterns better because:**

- Each plaintext block is XORed with the previous ciphertext block before encryption.
- Identical plaintext blocks produce different ciphertext values.
- It prevents attackers from recognizing repeated data patterns.
- ECB reveals repeated patterns, making it less secure for encrypting structured data such as images or documents.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation